

Predicting Customer Conversion in Digital Marketing

Capstone Three Project Report

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1. Problem Statement

Which customers are most likely to convert from a digital marketing campaign, and which demographic, behavioral, and campaign-level factors most strongly drive that conversion? The goal is a model that, given a customer's profile and the campaign targeting them, predicts conversion and surfaces the levers marketers can actually pull.

Marketing budgets face constant ROI scrutiny, and a large share of spend is wasted on customers who were never likely to convert. A reliable conversion model lets teams concentrate spend on high-probability segments, match customers to the channels and campaign types they respond to, and justify budget decisions with evidence rather than intuition.

2. Data

The dataset is a synthetic customer/campaign dataset from Kaggle (Rabie El Kharoua, “Predict Conversion in Digital Marketing”): 8,000 customer records across 20 columns spanning demographics, campaign attributes, engagement metrics, and loyalty indicators, with a binary Conversion target.

The data was clean going in — no nulls, no duplicates — but carried two important caveats:

- Heavy class imbalance: roughly 88% of records are positive (converted), leaving the minority non-converter class (12%) both harder to predict and more valuable to identify correctly.
- Two columns (AdvertisingPlatform, AdvertisingTool) were fully redacted and dropped as uninformative. The dataset also carries no timeframe, limiting temporal analysis.

A known limitation of the synthetic data surfaced during wrangling: some behavioral fields were internally inconsistent (e.g., zero website visits paired with a nonzero pages-per-visit value), which cannot occur in real traffic data. These fields were kept for modeling but treated with appropriate skepticism.

3. Exploratory Data Analysis

Correlation analysis showed weak collinearity between most features and the target, and no strong relationships among the categorical campaign/channel variables and conversion — somewhat concerning given the intuitive expectation that campaign type would matter more. A quick baseline logistic regression was run early to sanity-check whether any predictive signal existed at all.

That baseline (unweighted logistic regression) produced an ROC-AUC of 0.782, with ClickThroughRate emerging as the strongest single predictor, followed by TimeOnSite, EmailClicks, AdSpend, and EmailOpens — consistent with the intuition that prior engagement signals are the best predictor of future conversion. The model was, however, poor at identifying non-converters (class 0 precision of only 0.30), a direct consequence of the class imbalance.

4. Feature Engineering

Building on the EDA finding that engagement metrics were the strongest predictors, three engineered features were added ahead of modeling:

- EngagementScore — a composite pre-purchase engagement measure
- ClickToOpenRatio — email clicks relative to opens, as a quality-of-engagement signal
- EngagementPerVisit and AdSpendPerClick — normalized engagement and spend efficiency measures
- AgeGroup — age re-cast as a binned categorical rather than continuous, since age effects tend to be groupwise rather than linear

A correlation check confirmed EngagementScore in particular showed one of the strongest relationships with Conversion of any feature in the dataset, engineered or original.

5. Modeling Approach

5.1 Handling class imbalance

Two approaches were compared head-to-head on a logistic regression baseline: `class_weight='balanced'` versus SMOTE oversampling of the minority class (bringing the training set from 5,610 / 790 to a balanced 5,610 / 5,610). The two approaches produced nearly identical results (ROC-AUC 0.783 vs. 0.780), indicating the model had largely exhausted what could be gained from resampling alone.

5.2 Threshold tuning

Using the precision-recall curve for the SMOTE logistic regression model, the decision threshold was tuned to 0.70 (on class 0 probability) at the precision/recall crossover point. This shifted the tradeoff meaningfully — raising overall accuracy from 0.76 to 0.86 — though it did so partly by trading class-0 recall for precision, illustrating that accuracy alone is a misleading metric on this imbalanced target.

5.3 Model comparison

Beyond logistic regression, Gradient Boosting and an RBF-kernel SVM were trained on the SMOTE-resampled data to test whether a nonlinear model could better separate the classes.

Model	ROC-AUC	Accuracy	Class 0 F1	Class 1 F1
Log Reg (class-weighted)	0.783	0.75	0.42	0.84
Log Reg (SMOTE)	0.780	0.76	0.43	0.85
Log Reg (SMOTE, tuned threshold)	—	0.86	0.42	0.92
Gradient Boosting (SMOTE)	0.789	0.89	0.52	0.94
SVM — RBF (SMOTE)	0.726	0.83	0.40	0.90

Gradient Boosting produced the best results across the board: the highest ROC-AUC (0.789), the highest class-0 F1 (0.52), and the best overall accuracy (0.89). The SVM underperformed the linear baselines on ROC-AUC despite its nonlinear kernel.

6. Results and Model Selection

Gradient Boosting on SMOTE-balanced data was selected as the final model. It offers the best balance of identifying non-converters (the harder, more valuable class per the original project criteria) while maintaining strong performance on the majority class. That said, the improvement over the class-weighted logistic regression baseline was modest — an ROC-AUC gain of roughly 0.006 — which is an honest and important caveat rather than a resounding win.

This ceiling is consistent with what was observed in EDA: correlations between most features and the target were weak, and no amount of resampling or model complexity can manufacture signal that isn't in the underlying data. Given the dataset's heavy imbalance and the modest lift achievable, these results are in line with similar public benchmarks on this dataset.

7. Limitations and Future Work

- The data is synthetic; relationships may be cleaner and more separable than in a real campaign dataset, so generalization to live data is unproven.
- ConversionRate and ClickThroughRate carry some leakage risk as near-deterministic proxies for the target; results should be revisited with these more conservatively handled.
- With no timeframe field, temporal and platform-level analysis (e.g., campaign fatigue, day-of-week effects) was not possible.
- Future work could explore XGBoost/LightGBM with proper hyperparameter search, SHAP-based feature attribution for stakeholder-facing explanations, and cost-sensitive thresholding tied to actual marketing spend per customer rather than a generic precision/recall tradeoff.

8. Business Recommendation

For marketing stakeholders, the practical takeaway is that early engagement signals — click-through rate, time on site, email interaction, and prior purchase history — are far more predictive of conversion than demographic or campaign-channel attributes. Budget and targeting decisions should weight observed engagement behavior more heavily than audience segmentation by channel or campaign type alone.